BriVL-based audio representation: Data, Tasks, Baselines And Results

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***Abstract*—Recently, academic researchers and industry have gradually realized that the multimodal large model has obvious advantages in some performance and downstream tasks. The study of multimodal model is of great significance to promote the emergence of general artificial intelligence in the future. In this paper, we propose a robust audio representation learning method WavBriVL based on Bridging-Vision-and-Language (BriVL). WavBriVL projects audio, image and text into a shared embedded space, so that multi-modal applications can be realized. We also demonstrate the qualitative evaluation of the image generated from WavBriVL as a shared embedded space, with the main purposes of: (1) Develop a new useful/sable audio representation method; (2) Learn the correlation between audio and image; (3) Get the impression of AI when reading audio/music, that is, use audio to generate pictures. Experimental results show that this method can effectively affect downstream tasks, and it can effectively generate appropriate images from audio.**

***Index Terms*—Audio-Visual emotional impression, crossmodal matching, Multimodal Learning, Deep Neural Networks, Speech representation learning.**

I. Introduction

Sound and vision affect people's core cognition in many areas, such as feeling, information processing and communication. Sound and vision are closely related. However, most of the existing methods only have a single cognitive ability, and some only study text-vision, text-voice, etc. Now researchers have gradually realized that in some cases: (1) The self-supervised pre-training on large-scale Internet data is better than that of high-quality/manually labeled data sets[1], such as the recently amazing OpenAI-chatGPT[1] (2) Multimodal models are better than single or bimodal models[2][3], such as Microsoft's latest BEiT3[2] excellent multimodal models in all fields and tasks.

Data volume is the basic element for training large-scale language models. Since BERT[4] in 2018 (perhaps even earlier), the pre-training model of NLP has been benefiting from large-scale corpora. This paradigm has been used up to now. According to theory of OpenAI[5], the language model gradually reflects a scaling law (the rule that the model capacity increases with the model volume). In supervised learning, manual annotation of large amounts of data is very expensive, so self-supervised learning is valued for large model training. In order to expand the boundary of the research field and break the limitation of the lack of relevant resources[6], we explore a new multimodal self-monitoring model based on the latest excellent work: **Bridging-Vision-and-Language**[7].

[1]https://chat.openai.com/

[2]https://github.com/microsoft/unilm/tree/master/beit3

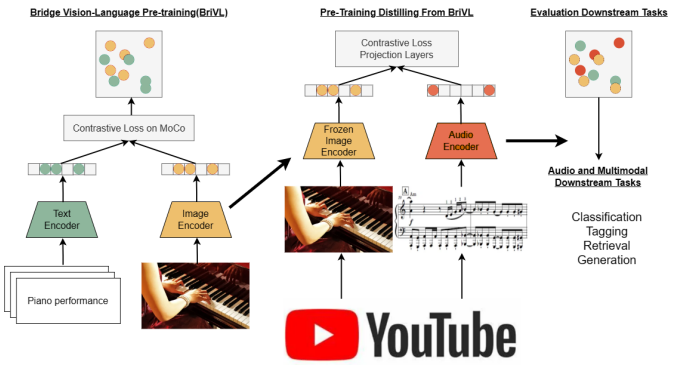

Fig. 1: Bridge Vidion-Language Pre-training (BriVL), and our two-stage approaches including pre-training and evaluation.

It is the first multi-modal large-scale pre-training model in Chinese. BriVL[3] model has excellent effect on image and text retrieval tasks, surpassing other common multimodal pre-training models (such as CLIP[4]) in the same period. In this work, we propose WavBriVL, an audio-visual correspondence model that extracts training from the BriVL model. The principle of WavBriVL is to freeze the BriVL visual model, run video on the visual stream of the model, and train a new model to predict BriVL embedding independently from the audio stream. Our method is very simple, which can train, expand, and output images. In addition, WavBriVL embeddings originate from BriVL, which means they align with text. In theory, this makes audio guided image repair[8], audio subtitles and cross mode text/audio to audio retrieval be true. And do some test for our work.

Finally, we use WavBriVL to guide the generation of model DALL·E-BriVL[5] output images, and intuitively verify that the embedded space is meaningful. Experimental results show that this method can effectively generate appropriate images from audio. In addition, compared with other fully supervised models, WavBriVL theoretically requires less data to obtain competitive performance in downstream tasks, that is, it performs pre-training more effectively than competitive methods, because it does not need to completely re learn the visual model, only needs to train the audio model. It is a reproducible and potential application model, and we will provide more code information after publication[6].

[3]https://github.com/BAAI-WuDao/BriVL

[4]https://github.com/openai/CLIP

[5]AnonymousGitHub

[6]AnonymousGitHub

II. METHOD AND TASKS

Bridging-Vision-and-Language(BriVL) is a model trained on 650 million text image weak semantic datasets. They designed a cross modal comparison learning algorithm based on the monomodal comparison learning method MoCo[9], and maintained the negative sample queue in different training batches through a mechanism called Momentum, so as to obtain a large number of negative samples for use in the comparison learning method. As shown in the left part of Figure 1, its core idea is to realize the general artificial intelligence model (AGI) by simulating the multi-mode processing idea of human brain. It also shows the SOTA results in such scenes as image annotation, image zero sample classification, and input features of other downstream multi-modal tasks. Even the guidance generation model has excellent performance.

As shown in Figure 1, WavBriVL replaces the text encoder with the audio encoder by freezing the visual model of BriVL, runs the image through it, and trains the new model to predict that only the matching image embedded content is obtained from the audio. We refer to the exclusive multilayer perceptron of BriVL, which can not only enhance performance but also prepare for possible downstream tasks in the future. After the audio encoder is trained, we freeze it and use it in the WavBriVL image generation task as a qualitative evaluation of our experimental results.

A. Dataset: Speech generation picture based on diffusion model

We used VGG-Sound[7] and AudioSet[8] video datasets. VGG-Sound consists of short clips of audio sound, which includes 310 video classes and 200,000 audio that span a large number of challenging acoustic environments and noise characteristics of practical applications. All videos are non man-made and more corresponding. We randomly select one image from each sample video, cut them into squares, and sample them down to 64 × 64. The audio sampling rate is 16,000Hz. We use it to train the model, which helps to increase the applicability of the model. AudioSet collected 10 second clips from 2.1 million videos. We randomly selected the audio of 18 videos for our image generation task.

B. Dataset: WavBriVL Performance test

We select diverse set of data ranging from various number of clips, number of categories, and perform diverse tasks including classification, retrieval, and generation. For evaluation, we use relevant metrics detailed in Table I for each task.

C. Feature Extraction Processing Methods

For image and audio encoders, we use EfficientNet-B7[10] as the CNN in the image encoder, and the backbone WavLM[11] as the basic transformer in the audio encoder. The self concerned block is composed of 4 Transformer encoder layers and MLP block respectively, with two fully connected layers and one ReLU activation layer. For all models, we use grid search to find the best hyperparameter. For other hyperparameters (such as batch size, training steps, learning rate, etc.), we directly use the suggested values in the original papers. Note that for per-instance perturbation, we adopt the appropriate quantity compared to the original epochs.

a) Image Encoder.: In the input image, the method of BriVL using random grayscale for the input image and random color jitter for data enhancement is followed. For all videos in the dataset, we use 720P resolution and separate images (if not, use 480P). All images are cropped down to 360 × 360 pixels. We use Transformer to capture patch features, and use the average pooling layer to fuse and extract. To better capture the relationship of image patch features, BriVL's team[9] deploy a self-attention (SA) block containing multiple Transformer encoder layers. Every Transformer encoder layer consists of a multi-head attention (MHA) layer and a feed forward network (FFN) layer[7]:

$$\mathbf{S}' = \text{LayerNorm}(\mathbf{S} + \text{MHA}(\mathbf{S})) \tag{1}$$

$$\mathbf{S} = \text{LayerNorm}(\mathbf{S}' + \text{FFN}(\mathbf{S}')) \tag{2}$$

Then, they use the average pooling layer to fuse the extracted patch features:

$$\mathbf{r}^{(i)} = \frac{1}{N_p} \sum_{j=1}^{N_p} \mathbf{S}_j \in \mathbb{R}^c \tag{3}$$

where $\mathbf{S}_j$ is the j-th column of $\mathbf{S}$. A two-layer MLP block with a ReLU activation layer is adopted to project $\mathbf{r}^{(i)}$ to the joint cross-modal embedding space, resulting in the final d-dimensional image embedding $\mathbf{z}^{(i)} \in \mathbb{R}^d$.

b) Audio Encoder.: For audio input, we first convert the original audio waveform (1D) into a spectrum (2D) as the input of WavLM, and pool the entire 512 dimensional audio sequence to output an embedding. The WavLM embedding is computed by the weighted average of outputs from all transformer layers. The WavLM[10] model inspired by HuBERT contains two main networks as follows: a CNN encoder and a Transformer with L blocks. During training, some frames of the CNN encoder output $\mathbf{x}$ are masked randomly and fed to the Transformer as input. The Transformer is optimized to predict the discrete target sequence $\mathbf{z}$, in which each $z_t \in [C]$ is a C-class categorical variable. The distribution over the classes is parameterized with

$$p(c|\mathbf{h}_t) = \frac{\exp(\text{sim}(\mathbf{W}^P \mathbf{h}_t^L, \mathbf{e}_c)/\tau)}{\sum_{c'=1}^{C} \exp(\text{sim}(\mathbf{W}^P \mathbf{h}_t^L, \mathbf{e}_{c'})/\tau)} \tag{4}$$

where $\mathbf{W}^P$ is a projection matrix, $\mathbf{h}_t^L$ is the output hidden state for step t, $\mathbf{e}_c$ is the embedding for class c, $\text{sim}(a, b)$ means the cosine similarity between a and b, and $\tau = 0.1$ scales the logit[11]. The WavLM embedding is calculated by the weighted average of all transformer layer outputs of WavLM, where the weights are learned during fine tuning. In the process of fine-tuning, we either update or freeze the parameters of WavLM.

[7]https://github.com/hche11/VGGSound
[8]https://research.google.com/audioset
[9]https://github.com/BAAI-WuDao/BriVL
[10]https://github.com/microsoft/unilm/tree/master/wavlm

D. Training process

We continue to use a similar cross modal comparative loss in BriVL. It is defined based on MoCo[9], which provides a mechanism of building dynamic sample queues for contrastive learning. Since the two negative queues used in our BriVL decouple the queue size from the mini-batch size, we can have a much larger negative sample size than the mini-batch size (thus GPU-resource-saving). Loss function with cross projection defined as $CXLoss = L(f(Image), Audio) + L(Image, g(Audio))$ (f, g: projection functions and L: contrastive loss).

III. Related Works

Our motivation comes from the relevant work proposed in the first half of this year (2022): we can see that BriVL has demonstrated better performance than CLIP[12] in many aspects, and Microsoft's new WavLM[11] is also better than the previous Wav2Vec[13] in most cases. We guess that the combination of these two new works will also be better than Wav2CLIP[11].

For many years, people have been trying to give AI people multimodal perception and thinking, and people's impressions of different external inputs are certainly a topic among them. With the emergence of different generation models, such as Goodfeed introduced GAN in 2014, there has been a lot of excellent work in the field of image generation. From single mode to multi-mode, from text guidance about 15 years later to audio guidance[14] 20 years later (of course, there are more and earlier attempts and exceptions), there are more than ten impressive works[15].

In addition to AudioCLIP and other similar but actually different work, the most similar is Wav2CLIP[16] Compared with their work, BriVL has more weak semantic relevance, so our model is more imaginative. Secondly, for our network architecture, because there is not necessarily a fine-grained area match between the image and audio, we lost the time-consuming target detector and adopted a simple and more efficient dual tower architecture, so we can encode the image and audio input through two independent encoders. Thirdly, BriVL designed a cross modal comparative learning algorithm based on the single modal comparative learning method MoCo[9], which has different advantages than CLIP.

IV. Task1: Speech generation picture based on diffusion model

To visually analyze the shared embedded space of WavBriVL and enhance its detectability. We asked WavBriVL to generate images with the help of DALL·E[12]. The purpose of this task is to qualitatively evaluate the effect of our model. The image of the previous model is generated by VQGAN, because the old model has not completely tried the effect of the diffusion model. The effect of their use of the diffusion model is worth looking forward to, but they have not yet tried it.

[11] https://github.com/descriptinc/lyrebird-wav2clip
[12] https://github.com/lucidrains/DALLE-pytorch

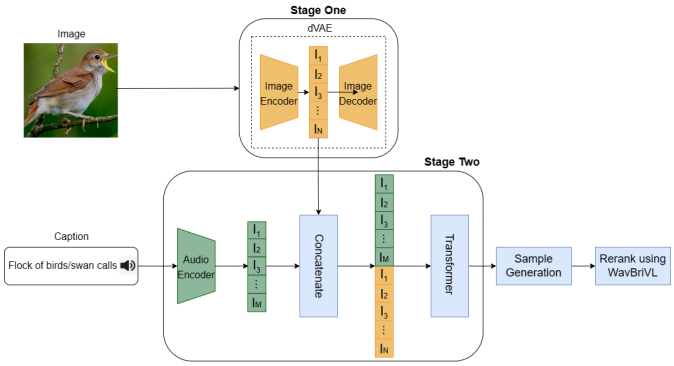

Fig. 2: Working principle of DALL·E+WavBriVL

A. Processing Method

WavBriVL includes an Audio Encoder and an Image Encoder. Its pre training model can accurately judge whether the given audio and image match. In order to generate audio guidance image, we only need to match the image generated by DALL·E with the input audio according to whether BriVL "approves" it. If there is no match, feedback will be given to DALL·E[17] to guide it to generate more accurate images. This process is an iterative process of training DALL·E. In this iteration process, the image quality generated by DALL·E must be getting better and better, and closer to the limit of WavBriVL.

DALL·E image generator was created by OpenAI and it can be able to generate images similar to surrealism directly through text description. VQGAN is the choice of Wav2CLIP, and not very convenient to compare similar work. But, comparisons are still relevant. The goal of DALL·E is to treat the text token and image token as a data sequence and carry out auto-regression through Transformer. Due to the large resolution of the image, if a single pixel is treated as a token, it will lead to a huge amount of computation, so DALL · E introduced a dVAE model to reduce the resolution of image. 1. In the first stage, first train a dVAE to compress each 256x256 RGB image into a 32x32 image token, and each position has 8192 possible values (that is, the encoder output of the dVAE is the logits with the dimension of 32x32x8192, and then combine the features of the codebook through the logits index. The embedding of the codebook is learnable). 2. In the second stage, the text is encoded with WavLM Encoder to obtain a maximum of x audio tokens. If the number of tokens does not meet the maximum value, the maximum value is padded. Then x text tokens and 1024 image tokens are spliced to obtain 1280 data in length. Finally, the spliced data is input into Transformer for autoregressive training. 3. In the reasoning stage, given a candidate image and an audio, the fused token can be obtained through the transformer, and then the image can be generated by the dvae decoder. Finally, the matching score of the audio and the generated image can be calculated through the pre-trained WavBriVL. As in general performance testing, DALL·E and WavBriVL are frozen during the generation process.

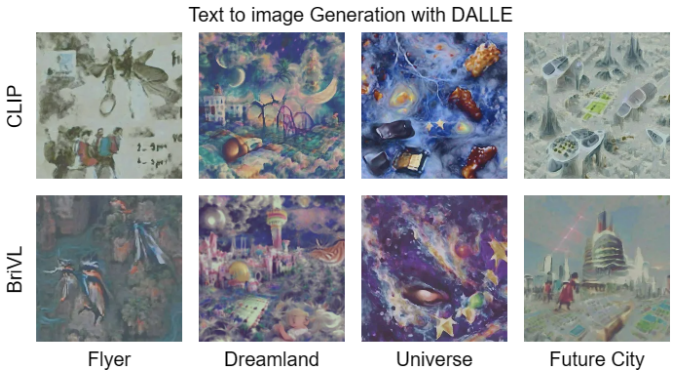

Fig. 3: Examples of CLIP (top) and BriVL (bottom) to image generation from AudioSet, corresponding labels (translated) in x-axis.

Fig. 4: Images generated from five audio in AudioSet [18]. Top: Wav2CLIP, Bottom: WavBriVL - corresponding labels in x-axis.

B. Comparison with previous work

In previous work, Wav2CLIP also tried to generate text/audio maps. Here are two maps for comparison with our work. Figure 3 shows the text output image of CLIP and BriVL. Figure 4 shows another group of pictures generated by Wav2CLIP and WavBriVL using audio.

We can see that the style of text generation diagram is similar, which is because they all use the same generation model. However, in details, they have their own characteristics: for example, in their understanding of "Flyer", English has both the meaning of flying objects and flyers, while Chinese is closer to the meaning of birds. If it is a translation difference, then the login elements and magpies can show that BriVL has more Chinese characteristics, and the rest of the pictures have similar characteristics;

For the generation of audio, they show two characteristics of convergence and divergence between the two models, of course, the images are still corresponding as we see them. The convergence is shown in the similarity of images, which is because our training set is English images that weaken the style of BriVL. Divergence is shown in Figure 4, which is more imaginative than that in Figure 3. This is because BriVL's weak semantic text image data set has a strong imagination, and the other is because the audio itself has a strong divergent ability, which will enhance the model's associative ability.

V. TASK 2: WAVBRIVL PERFORMANCE TEST

A. Training, development, and evaluation sets

We selected publicly available audio classification data of different sizes, which are generally used for evaluation[19], and also included some audio tasks/data, as shown in the table I, including classification, retrieval and audio catalogue. ESC-50[20] is a simple data set with only 2 thousand samples, while UrbanSound8K[21] is a large environmental data set with 10 categories. VGGSound[22] is a huge set of audio and video materials, including the widest, widest and widest range of audio molds. DESED is used again as an audio extraction (AR) job because DESED can perform sound extraction at the fragment level. Finally, Clotho[23] is a unique set of audio subtitles.

Dataset	Task	Clip (Split)	Class	Metric
ESC-50 [20]	MC/ZS	2k (5 folds)	50	ACC
UrbanSound8K [21]	MC/ZS	8k (10 folds)	10	ACC
VGGSound [22]	MC/ZS	185k	309	mAP
DESED [24]	AR	2.5k (valid)	10	F1
VGGSound [22]	CMR	15k (test)	309	MRR
Clotho [23]	AC	5k (evaluation)		COCO[13]

TABLE I: Downstream tasks, including 1. classification: multi-class (MC), zero-shot (ZS), 2. retrieval: audio (AR) and cross-modal retrieval (CMR), and 3. audio captioning (AC) task, with various of clips, of classes, and common metrics.

For multi-class (MC) classification problems, an MLP-based classifier is employed, with a corresponding number of classes as output. In DESED, we use the way of simulating WavBriVL and sed_eval[14] to realize audio retrieval (AR). At the same time, we also explore the performance of Wav2BriVL when dealing with multimodal tasks, and how to transfer zero samples to other modalities.

B. Comparisons with previous work

First, we monitor the benchmark by training from scratch on each downlink (with random initialization of the encoder weights). Next, we compare WavBriVL with other publicly available OpenL3[19] pre-trained on different pretext tasks in OpenL3. OpenL3 multimodal self-monitoring training with AudioSet. It serves as a strong benchmark for different audio tasks, such as audio classification and retrieval. We extract features from OpenL3 (512 dim) and WavBriVL (512 dim) and apply the same training scheme to all downstream classification and retrieval tasks.

It can be noted that our model has good effects in both data sets of audio retrieval classification, one of which may be related to our data, and the other may be the effect of BriVL. In the mrr tasks from A to I, we have also achieved excellent results, which mainly comes from the excellent training effect of the previous two towers model and the pre-training model, the structure of the brief is useful for coping with tasks.

[14]https://github.com/TUT-ARG/sed_eval

Model	Classification				Retrieval	
	ESC-50	UrbanSound8K	VGGSound	DESED (AR)	VGGSound (CMR)	
	ACC	ACC	mAP	F1	A→I (MRR)	I→A (MRR)
Supervise	0.5200	0.6179	0.4331			
OpenL3	0.733	0.7588	0.3487	0.1170	0.0169	0.0162
Wav2CLIP	0.8595	0.8101	0.4663	0.3955	0.0566	0.0678
WavBriVL	**0.8617**	0.7832	**0.4741**	0.3420	**0.0611**	0.0608
SOTA	0.959 [25]	0.8949 [25]	0.544 [26]			
WavBriVL (ZS)	0.412	0.4024	0.1001			

TABLE II: In the subsequent classification and acquisition work, there will be supervised training, other audio representation modes, OpenL3, and the latest SOTA. ZS is based on WavBriVL as a zero sample size model, some of which are derived from the original literature.

VI. SUMMARY & CONCLUSIONS

This paper introduces a WavBriVL[15] for audio representation. The results show that WavBriVL is able to output general, robust sound representations, and that WavBriVL can be easily transferred to multimodal jobs, such as audio generation diagrams. In future research, we will explore an interpretable machine learning approach that uses the ability to generate (sound-image) across modalities. Based on learnings from embedded systems, additional speech classification and retrieval efforts are evaluated and compared to more advanced speech models. On this basis, we will try to share the embedding space in multiple modes, so as to achieve the cross mode of image-generated text and image-generated sound. In the future, we will also consider using Microsoft's latest text-to-speech fusion model, SpeechLM[27], and the next release of the Diffusion model[28], as the next version of the work.

LIMITATIONS

The limitation of WavBriVL is that the BriVL is trained based on the Chinese text image data set WSCD (the text-image is corresponding; [7]), while our later training uses the English video data set VGG-Sound (the audio-image is corresponding). However, audio and text do not necessarily correspond strictly. It has no impact on the classification, retrieval, and generation tasks of audio image, but it is not recommended to use them when text image tasks are involved (of course, this is obviously a BriVL task, not our WavBriVL task). When future researchers explore multimodal mutual transformation, it is recommended to find a Chinese video dataset for retraining.

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